BIG DATA – BIG DATA ANALYTICS

Written exam DEMO 2026

1. Given the `caesarian.csv` dataset, which contains information about caesarian section results of 80 pregnant women with the most important characteristics of delivery problems in the medical field, namely age, delivery number (number of previous pregnancies), delivery time, blood pressure and heart status, and the label assessing if the delivery was caesarian or not.
    - Create a custom train-test split with 90% as training set and the remaining 10% as test set;
    - Implement a strategy to handle missing data and explain your reasoning;
    - Implement a strategy to convert the categorical data (also the label value) into a meaningful numeric form;
    - Apply the so obtained transformation to the test set.

points ___/ 8

2. Use the cured dataset to train a `RandomForestClassifier` with the following the given hyperparameter grid:
    - Split criterion: Gini, log_loss
    - Number of trees: 100, 150
    - Maximum depth of trees: 2, 5, 10

    Consider the accuracy as metric to select the best model over the training set, then print the best combination of the found hyperparameter.

points ___/ 9

3. Implement a PyTorch neural network with at least one hidden layer, with a dropout layer with dropout=0.2. Train the network for at most 30 epochs with SGD optimizer, with momentum of 0.9 and weight decay equal to 0.1. Apply early stopping over the validation set (10% of the training set) with a patient of 5 epochs and a minimum increment of 0.01

points ___/ 10

4. Calculate the accuracy over the test set with the two trained model, print the complete classification report and the confusion matrix.

points ___/ 3

Total: points ___/ 30

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Written exam rules

The following are the rules to follow and the characteristics of the exam for evaluation purposes:

1. The total duration of the exam is two hours and includes a series of questions that explore different aspects of the same problem: everyone will be free to dedicate the time they wish to each question.
2. The exam is conducted entirely on a computer.
3. The prepared development environment is a pip environment called `bigdata-env`, directly accessible from Visual Studio Code.
4. To start the environment, open Visual Studio Code and, from its file manager, create a new folder named NameSurname on the Desktop and create a new notebook for which the previously illustrated environment will be selected.
5. Submission of the exam will be permitted after the first hour of the exam.
6. It will be necessary to switch off and hand in mobile devices (smartphones, smartwatches, and tablets) to the lecturer before the start of the exam.
7. Internet navigation from the workstations will be blocked, in general, and permitted only towards the documentation websites of the libraries and the PDF slide repository.
8. The lecturer will distribute a digital copy of the assignment and any data sets directly from their workstation or via a USB drive, and will collect the programming work in the same manner.
9. The candidate must, in any case, hand in this sheet, dated and including their name and student ID number.
10. For the purpose of calculating the final grade, the *maximum value* for each question is indicated at the bottom of the question itself. The exam will receive a grade equal to the sum of the scores obtained for each question. It is specified that the lecturer will assign a *non-binary score* to each question—that is, not just the full value or 0—but will evaluate the formal correctness of the work, the methodological rigor of the theoretical approach, and the originality of the proposed solutions to assign a score within the range defined by the question's maximum value.

Student: ______________________________ Student ID Number: ____________